Splunk-based Security Operations Center (SOC) Dashboard

Summer Internship Project Report

A Project Report

Submitted in fulfilment of the Summer Internship Programme

B.Tech Computer Science and Engineering (Cyber Security & Digital Forensics)



Abstract

This project presents an AI-assisted Security Information and Event Management (SIEM) dashboard developed using Splunk Enterprise. The system collects Linux authentication logs including SSH and sudo activities, indexes them within Splunk, and analyzes security events using Search Processing Language (SPL). Interactive dashboards display key performance indicators, event timelines, recent authentication events, and MITRE ATT&CK mappings. The platform also generates an AI-assisted summary of indexed events to help analysts quickly understand the overall security posture. The project demonstrates how a Security Operations Center (SOC) can monitor, visualize, and investigate security events using Splunk Enterprise.

Introduction

Splunk Enterprise is one of the industry's most widely adopted SIEM platforms because of its powerful log indexing, Search Processing Language (SPL), interactive dashboards, and real-time monitoring capabilities. This project demonstrates these core SIEM concepts using Linux authentication logs and MITRE ATT&CK mapping.

Problem Statement

Security teams are often overwhelmed by the sheer volume of logs generated across an organization's infrastructure. Authentication systems, firewalls, intrusion detection systems, and operating systems each produce their own log formats, and reviewing them manually is neither practical nor scalable. As a result, genuine threats can go unnoticed among routine log entries, and by the time an incident is discovered, significant damage may already have occurred.

Purely rule-based detection systems, while useful, are limited to identifying attack patterns that have already been defined in advance. They cannot easily adapt to new or unusual behaviour that does not match an existing rule. On the other hand, analyzing logs manually, without any structured pipeline or centralized dashboard, does not scale beyond a very small environment. There is a clear need for a system that can:

- Centralize Linux authentication logs.
- Monitor SSH and sudo activities.
- Detect suspicious authentication attempts.
- Visualize security events.
- Assist SOC analysts.

Objectives

- Configure Splunk Enterprise.
- Monitor Linux authentication logs.
- Build SPL searches.
- Develop interactive dashboards.
- Map detections to MITRE ATT&CK.
- Generate AI-assisted summaries.
- Demonstrate SOC monitoring.

Dataset Description

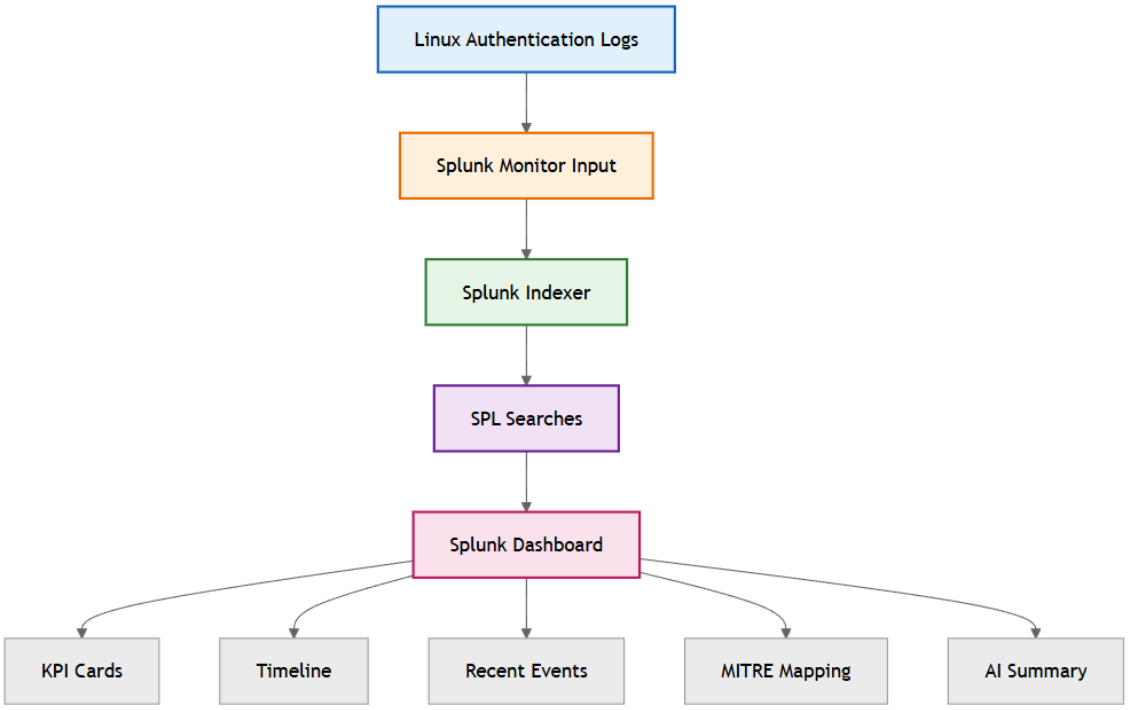
The SIEM dashboard processes Linux security logs generated from authentication services, SSH sessions, sudo activity, and system logs. These logs are indexed by Splunk Enterprise and analyzed using Search Processing Language (SPL). Instead of relying on a traditional dashboard components and SPL searches dataset, the system operates directly on real security event logs collected from the monitored environment.

Log Source	Representative Fields
Authentication Logs	SSH Authentication
sudo Logs	Privilege Escalation
SSH logs	Login Monitoring
linux Syslog	System Events

During preprocessing, these differently structured logs are mapped onto a common internal schema so that downstream components, such as the correlation engine and the dashboard components and SPL searches , can operate on a consistent set of fields regardless of the original log source. Fields that are not relevant to detection, such as free-text descriptions, are filtered out, while fields useful for identifying patterns, such as source IP, timestamp, and event type, are retained and standardized.

System Architecture Design

Splunk-based SIEM Dashboard follows a layered architecture that separates log ingestion, detection, storage, and presentation into distinct stages. This separation keeps each part of the system independent, so that the detection logic, for example, can be modified without affecting how logs are collected or how the dashboard displays results. The diagram below shows the overall flow of data through the system, from raw log sources to the final Security Operations Center dashboard.



Module Descriptions

Splunk Monitor Input: Collects Linux log files.

Splunk Indexer: Indexes logs.

SPL Searches: Analyzes logs.

Dashboard: Visualizes security events.

MITRE Mapping: Maps detections.

AI Summary: Summarizes indexed events.

Methodology

The development of the SIEM Analytics System followed a structured methodology, moving from raw log ingestion through log ingestion, indexing, SPL-based analysis, dashboard visualization and MITRE ATT&CK mapping, before finally presenting results on the dashboard. The table below summarizes the method used at each stage of the pipeline.

Stage	Tool
Log Monitoring	Splunk Monitor Input
Log Indexing	Splunk Enterprise
Searching	SPL
Dashboard	Simple XML
MITRE Mapping	SPL
Visualization	Splunk Dashboard

Data Collection

Splunk Monitor Input continuously watches Linux authentication log files and indexes newly generated events automatically.

Data Preprocessing

During indexing, Splunk automatically extracts fields such as timestamp, hostname, source type, username, process information, and authentication status. These fields are used by SPL queries to generate dashboard visualizations and security metrics. .

Dashboard

The Security Operations Center dashboard is the primary interface through which analysts interact with the system. It contains the following components:

- KPI Cards
- Event Timeline
- Recent Events
- MITRE ATT&CK Mapping
- AI Summary

All dashboard panels are dynamically updated using Search Processing Language (SPL) queries executed on indexed log data. This ensures that the displayed information accurately reflects the current security events within the monitored environment.

Technology Stack

The system uses a compact and widely adopted technology stack, chosen to keep the project practical to build, run, and extend within the scope of a university project.

Technology	Purpose
Splunk Enterprise	SIEM
Linux	Log Source
Bash	Test Log Generation
SPL	Search
Simple XML	Dashboard
MITRE ATT&CK	Threat Mapping

Key Features

- Live KPI Cards
- Event Timeline
- Recent Events Table
- MITRE ATT&CK Mapping
- AI Summary
- Linux Authentication Monitoring
- Dynamic SPL Searches

Project Modules

Module	Description
Monitor Input	Watches and ingests raw Linux authentication logs
Indexing	Processes and stores logs in Splunk Enterprise
SPL Search	Analyzes and extracts meaningful insights from indexed logs
Dashboard	Visualizes security events through interactive panels
MITRE Mapping	Maps detected events to MITRE ATT&CK techniques
AI Summary	Generates dynamic summaries of security activity

Advantages

- Centralized Monitoring
- Dynamic Dashboards
- MITRE Mapping
- SPL Searching
- AI Summary
- Easy Deployment
- Offers a clear, visual dashboard that makes security monitoring accessible even to users who are not deeply technical.
- Demonstrates practical Security Operations Center (SOC) monitoring using Splunk Enterprise.

Limitations

- Linux logs only
- Limited detection rules
- No threat intelligence
- No geographic attack map
- No automated risk scoring
- Educational platform

Expected Outcomes

- Live Dashboard
- Authentication Monitoring
- MITRE Mapping
- AI Summary
- Interactive Visualization

Applications

- SOC Training
- Cybersecurity Education
- Linux Log Monitoring
- SIEM Demonstration
- Splunk Learning

Conclusion

The proposed project demonstrates how Splunk Enterprise can be used to build an educational Security Information and Event Management (SIEM) dashboard capable of collecting Linux authentication logs, executing SPL-based searches, mapping detections to the MITRE ATT&CK framework, and presenting results through interactive dashboards. The implementation provides practical exposure to SOC monitoring workflows and illustrates the fundamental concepts of centralized security event management.

The project also provides practical experience in configuring Splunk Enterprise, writing SPL queries, monitoring Linux authentication logs, and understanding Security Operations Center (SOC) workflows. It serves as a strong educational platform for learning modern SIEM concepts and cybersecurity monitoring techniques.

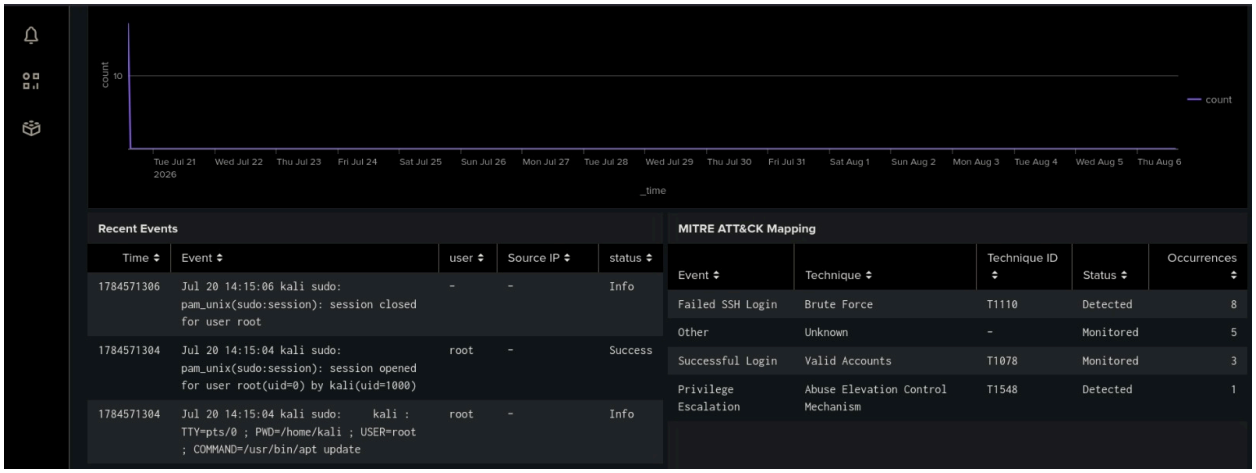
Future Scope

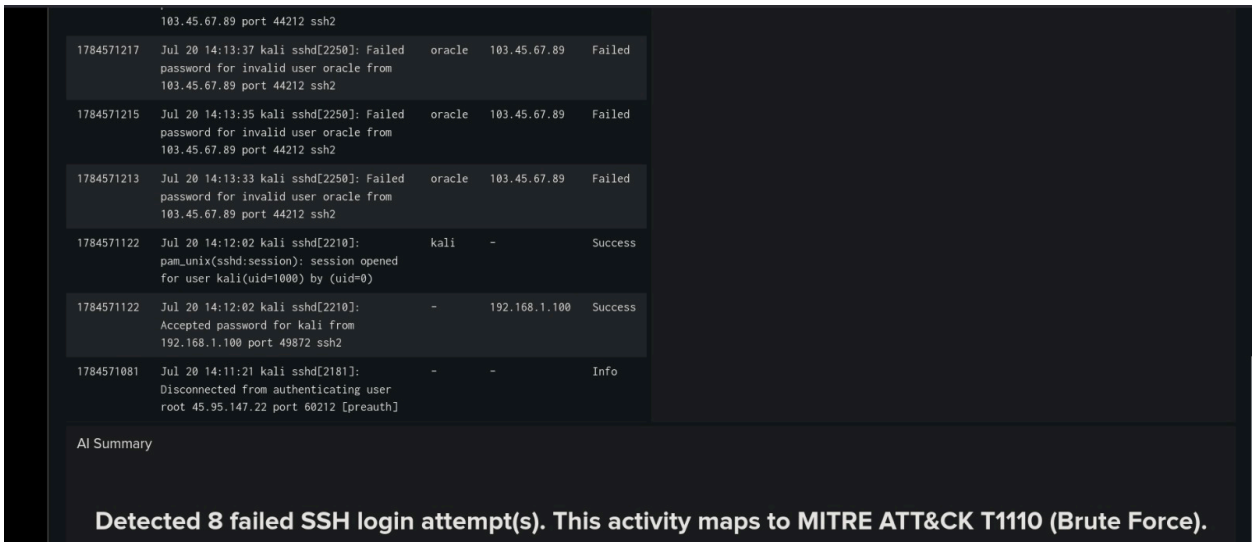
- Live Log Ingestion
- Detection Rules & Alerts
- Threat Intelligence Feed
- MITRE Heatmap
- MITRE-ATLAS Mapping
- Risk Score Calculation
- Geographic Attack Map

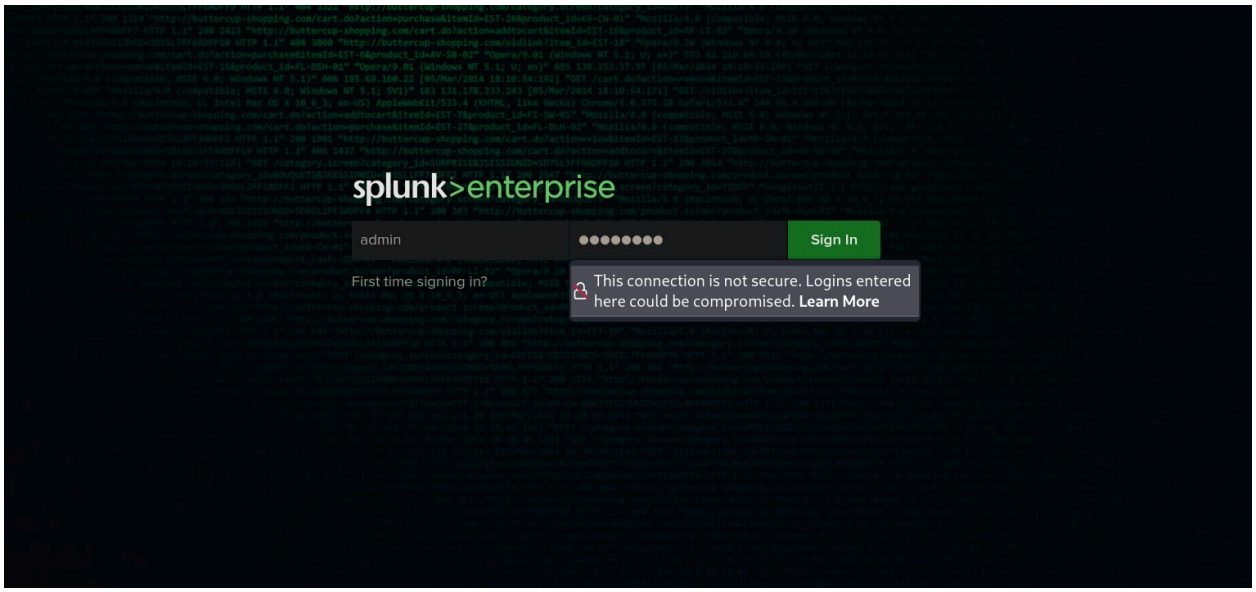
References

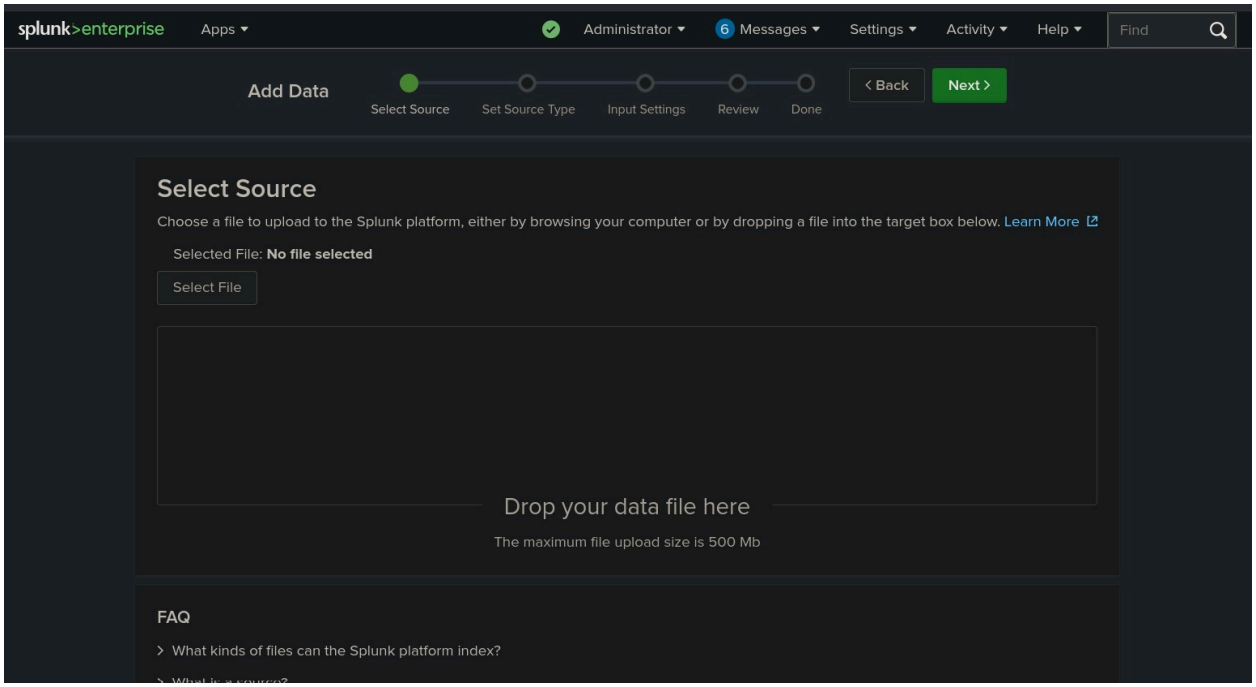
- Splunk Enterprise Documentation
- Splunk Search Processing Language (SPL) Documentation
- MITRE ATT&CK Framework
- NIST SP 800-61 Revision 2: Computer Security Incident Handling Guide
- OWASP Logging Cheat Sheet
- RFC 5424 – The Syslog Protocol

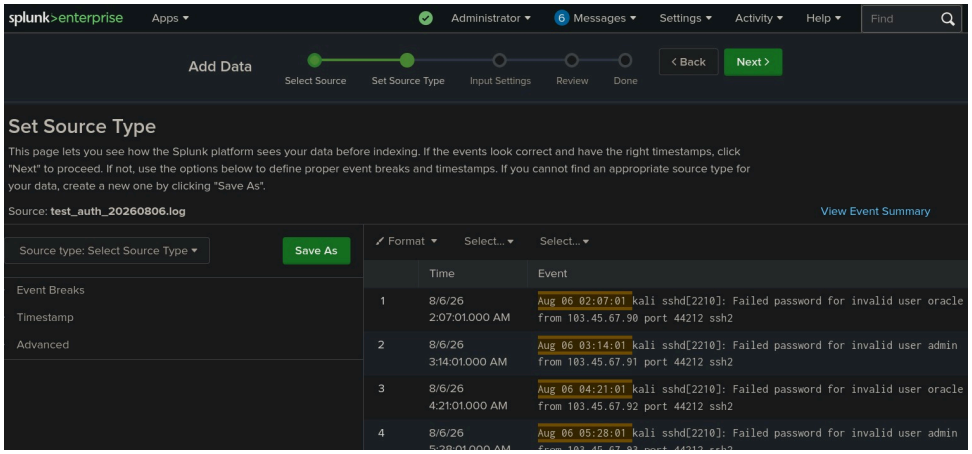
Screenshots

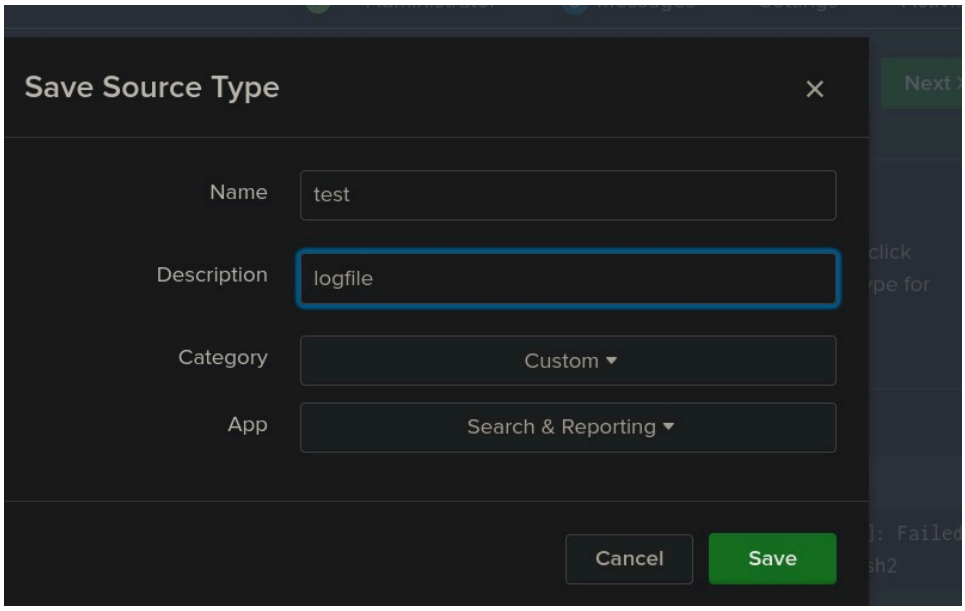


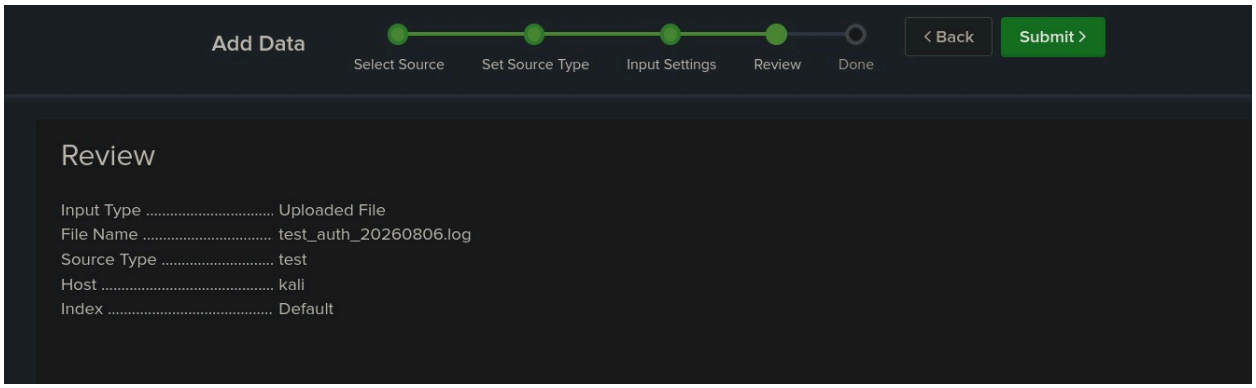



splunk>enterprise
admin
Sign In
First time signing in?
This connection is not secure. Logins entered here could be compromised. Learn More


splunk>enterprise
Apps
Administrator
6 Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
< Back
Next >
Select Source
Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. Learn More
Selected File: No file selected
Select File
Drop your data file here
The maximum file upload size is 500 Mb
FAQ
What kinds of files can the Splunk platform index?


splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Next
Set Source Type
This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".
Source: test_auth_20260806.log
View Event Summary
Source type: Select Source Type
Save As
Event Breaks
Timestamp
Advanced
Format
Select...
Select...
Time
Event
1
8/6/26 2:07:01.000 AM
Aug 06 02:07:01 kali sshd[2210]: Failed password for invalid user oracle from 103.45.67.90 port 44212 ssh2
2
8/6/26 3:14:01.000 AM
Aug 06 03:14:01 kali sshd[2210]: Failed password for invalid user admin from 103.45.67.91 port 44212 ssh2
3
8/6/26 4:21:01.000 AM
Aug 06 04:21:01 kali sshd[2210]: Failed password for invalid user oracle from 103.45.67.92 port 44212 ssh2
4
8/6/26
Aug 06 05:28:01 kali sshd[2210]: Failed password for invalid user admin


Save Source Type
Name
test
Description
logfile
Category
Custom
App
Search & Reporting
Cancel
Save


Add Data
Select Source
Set Source Type
Input Settings
Review
Done
Back
Submit
Review
Input Type ..... Uploaded File
File Name ..... test_auth_20260806.log
Source Type ..... test
Host ..... kali
Index ..... Default